ENTERPRISE SYSTEM SPECIFICATION • VERSION 2.4

CURATION KITCHEN

The Smart Digital Culinary Assistant

COMPREHENSIVE TECHNICAL & ARCHITECTURAL APP OVERVIEW

An end-to-end architectural evaluation covering full-stack system topology, cross-platform clients (React 19 & Native Android), Gemini AI reasoning pipeline, Health Canada DRI baseline nutrition intelligence, e-commerce grocery fulfillment integration, and monetization framework.

DOCUMENT METADATA

Project Name:	**Curation Kitchen**	Runtime Engine:	**Node.js v22 / Express v4.21**
Target Architecture:	**Web (React 19) + Android (Compose)**	AI Foundation:	**Google Gemini 3.6 / 2.5 Flash**
Nutritional Baseline:	**Health Canada DRIs (Default)**	Fulfillment Matrix:	**Instacart, Walmart, Voilà**
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EXECUTIVE BRIEFING & SYSTEM INTENT

Curation Kitchen resolves the fragmentation between culinary inspiration, evidence-based nutrition science, household pantry management, and grocery logistics. Built upon a unified TypeScript full-stack core and accompanied by a native Jetpack Compose mobile client, Curation Kitchen enforces dietary confidence by defaulting to Health Canada's Dietary Reference Intakes (DRIs) on launch while supporting seamless transitions to United States (USDA) and International (WHO) dietary standards.

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1.0 EXECUTIVE OVERVIEW & PROBLEM SPACE

The Friction in Modern Culinary Wellness

The digital cooking space has historically suffered from extreme domain fragmentation. Home cooks routinely oscillate between fragmented recipe discovery blogs burdened with advertisement clutter, disconnected manual grocery lists, ungrounded calorie tracking calculators, and separate retail e-commerce applications. This disjointed experience results in substantial food waste, miscalculated dietary intake, and abandoned home meal routines.

Curation Kitchen introduces a tightly coupled, intelligent culinary operating system. The platform orchestrates four fundamental operational loops into a single cohesive feedback cycle:

- **Inspiration to Ingestion:** AI-powered extraction parses raw URLs, messy text, and voice dictations into normalized, strongly-typed recipe data models with zero formatting noise.
- **Stock to Recipe Alignment:** Live pantry tracking automatically informs meal feasibility, instantly identifying missing ingredients before cooking begins.
- **Evidence-Based Nutrition:** Eliminating arbitrary diet fads by standardizing daily targets against official public health baselines, defaulting to Health Canada DRIs on boot.
- **Frictionless Fulfillment:** Deep-linked, localized e-commerce routing for immediate grocery ordering with pre-configured Canadian fulfillment partner integration.

2.0 FULL-STACK SYSTEM ARCHITECTURE

System Topology & Cross-Platform Blueprint

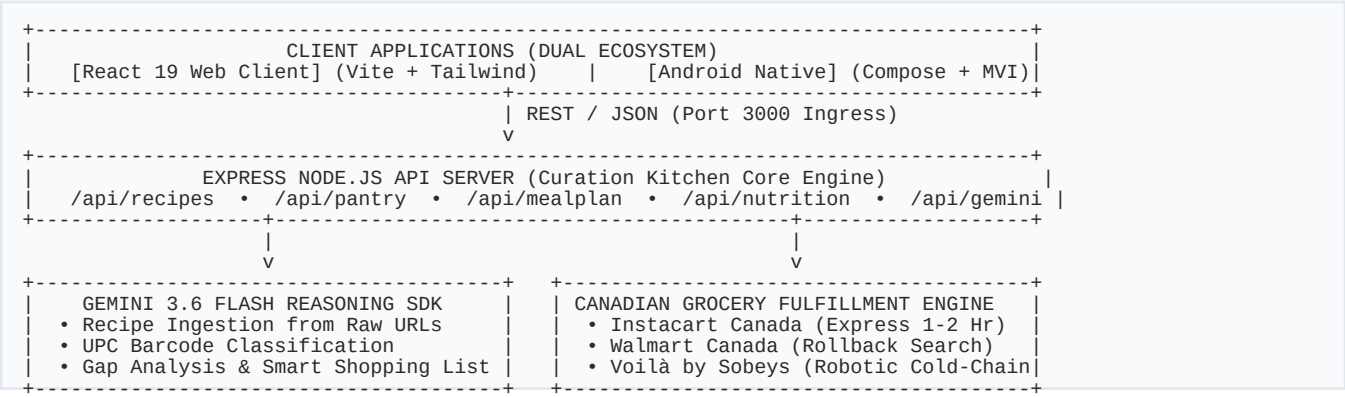
Curation Kitchen adheres to a server-authoritative, client-resilient architecture designed to operate smoothly across diverse device constraints, unstable mobile network environments, and sandboxed cloud containers.

ARCHITECTURAL TOPOLOGY OVERVIEW

- Client Layer: React 19 SPA (Web) + Jetpack Compose Native App (Android)
- API Gateway & Middleware: Express 4.21 on Node.js 22 with Vite SSR/SPA middleware
- AI Intelligence: Google Gemini 3.6 Flash / 2.5 Flash via Server-Side SDK (@google/genai)
- Local & Remote Persistence: JSON Document Store (db.json) with ACID write guarantees, SQLite/Cloud SQL read models, and Android Room Local DB
- Fulfillment Engine: Automated affiliate-encoded deep link generator for Instacart, Walmart, and Voilà

2.1 High-Level Component Interactions

The diagram below reflects the architectural data flow governing client synchronization, artificial intelligence grounding, and external fulfillment dispatch:



2.2 Data Integrity & Sandboxed Execution

The backend engine operates inside sandboxed cloud environments where external ingress is strictly bound to Port 3000 behind reverse-proxy routing. In compliance with security standards, all Gemini API tokens are strictly isolated in server memory and never broadcast to client user agents. Development assets are bound via Vite middleware in SPA mode, seamlessly compiling to standalone CommonJS bundles for production deployment.

3.0 WEB CLIENT ARCHITECTURE

React 19, Modern Tailwind CSS, & Workspace Grid

The web application is engineered as an ultra-responsive, zero-latency Single Page Application (SPA) leveraging the latest features of React 19, Tailwind CSS v4, and TanStack Query v5. The user interface eschews generic dashboard templates in favor of a cohesive, human-centered culinary aesthetic.

3.1 The Culinary Workspace Grid Layout

Rather than burying essential kitchen utilities behind multi-nested drawer menus, Curation Kitchen features the Culinary Workspace Grid. Configured as a balanced 2x2 (mobile) and 3x2 (tablet/desktop) matrix, the workspace tiles provide immediate, tactile access to high-frequency kitchen operations:

Workspace Tile	Primary Function	Key Interaction State
Pantry Hub	Ingredient ledger & stock health	Live badge count, threshold alerts
Meal Planner	7-Day calorie & macro calendar	Slot allocation, multi-day scheduling
Collections	Thematic cookbooks & folders	Color-coded tags, recipe curation
Categories	Supermarket department organizer	Ingredient grouping, pantry assignment
UPC Scanner	Live camera barcode recognition	Open Food Facts lookup & AI parsing
Local Search	Instantaneous client-side search	Full-text indexing across titles/tags

3.2 Design System & Visual Identity

The visual language embodies a 'Quiet Warm Luxury' ethos tailored specifically for cooking environments where visual clarity, readability at arm's length, and warm neutral tones prevent eye fatigue:

- **Buttercream Canvas (#FAF6EE):** An organic, warm cream background replacing sterile hospital-white canvases, providing high visual comfort in varied ambient kitchen lighting.
- **Terracotta Orange (#D95338):** The primary operational accent color, representing warmth, culinary heat, and decisive user actions (import, save, order delivery).
- **Sage Green (#6E8B6B):** The secondary harmony tone, utilized for wholesome nutritional indicators, 'In Stock' pantry verifications, and balanced macro states.
- **Charcoal Typography (#2C2A29):** High-contrast, deep slate body text passing WCAG AAA contrast requirements without the jarring harshness of pure #000000.

3.3 State Management & Data Synchronization

Server state is orchestrated exclusively via TanStack Query (React Query v5). Queries enforce optimistic updates and stale-while-revalidate caching policies. When a user marks an ingredient as depleted or adjusts a meal slot, the UI immediately transitions to the desired state while executing background asynchronous persistence, ensuring an effortless, lag-free user experience.

4.0 NATIVE MOBILE CLIENT ARCHITECTURE

Jetpack Compose, Android Architecture Components, & MVI

To guarantee superior ergonomics in active cooking scenarios, Curation Kitchen includes a native Android mobile client built in modern Kotlin with Jetpack Compose, Material 3, and a Unidirectional Data Flow (MVI) architecture.

4.1 Architecture Components & Layering

- **Presentation Layer (Compose UI):** Declarative, reactive UI composables utilizing Material 3 design tokens customized to match the Buttercream (#FAF6EE) and Terracotta (#D95338) visual system. Screens react to immutable UI state exposed via StateFlow.
- **State Management & ViewModels:** Architecture-compliant ViewModels encapsulate business logic, managing UI events (e.g. barcode scan, quantity decrement, meal drag) and transforming repository data streams into clean ViewStates.
- **Repository Pattern:** The PantryRepository and RecipeRepository abstract the underlying data sources, synchronizing with the backend REST API via OkHttp/Retrofit while caching locally for offline availability.
- **Preferences DataStore:** User preferences, dietary flags, and the regional baseline selection (Health Canada DRI vs. USDA vs. WHO) are persisted locally via Jetpack DataStore.

4.2 Mobile Screen Modules & Workflows

Mobile Screen	Technical File	Key Capabilities
Home Screen	HomeScreen.kt	Hero culinary banner, quick action bar, recommended recipes
Pantry Hub	PantryScreen.kt	Department list, float quantity +/- stepper, barcode triggers
Meal Planner	MealPlannerScreen.kt	Day-by-day swipe carousel, slot assignment, calorie meters
Nutrition Hub	NutritionScreen.kt	Circular progress dials, Health Canada DRI vs USDA toggle
Paywall Screen	PaywallScreen.kt	Google Play Billing, tier tiers, promotional feature grid

4.3 In-App Billing & Monetization Gatekeeper

The Android client integrates Google Play Billing through a dedicated BillingRepository and PremiumGatekeeper. While the foundational cooking utilities (recipe storage, local pantry, basic nutrition) remain completely free, advanced automation features (unlimited AI URL ingestion, predictive meal planning, multi-household sync) are gracefully gated behind subscription tiers (Monthly Chef & Annual Culinary Master).

5.0 CORE FUNCTIONAL DOMAINS & WORKFLOWS

Modular Kitchen Capabilities & Operational Depth

Curation Kitchen decomposes culinary management into five tightly integrated functional domains. Each domain addresses a critical stage of the household food lifecycle.

5.1 Domain A: Intelligent Recipe Ingestion & Intelligence

The recipe ingestion pipeline accepts raw inputs from two primary modalities: public web recipe URLs and unstructured text/dictation. The backend normalizes this input into a standardized recipe schema containing yield, prep time, cook time, granular ingredient lists (with numerical quantity, unit, and item name), step-by-step instructions, and nutrient estimates.

5.2 Domain B: Intelligent Pantry Management & Shelf Hub

The pantry system treats kitchen inventory as a real-time ledger. Key capabilities include:

- **Floating-Point Precision:** Inventory units are tracked as clean decimal floats (e.g., 2.5 cups of flour, 0.75 kg of rice), accommodating partial uses and realistic kitchen measurements.
- **Taxonomic Department Classification:** Ingredients are automatically categorized into Produce, Dairy & Refrigerated, Baking & Spices, Grains & Pasta, Seafood & Proteins, and Pantry Essentials.
- **Depletion & Restock Workflows:** When ingredients reach zero, the item automatically flags as 'Out of Stock / Need to Buy', immediately surfacing in the smart shopping generator and delivery modal.
- **UPC Barcode Recognition:** Integrated HTML5 barcode scanning leverages device cameras to decode UPC/EAN barcodes, fetching product data and cross-referencing Open Food Facts.

5.3 Domain C: 7-Day Meal Planner & Dynamic Slot Allocation

The weekly meal planner organizes nutrition over rolling 7-day windows. Each calendar day provides distinct meal slots (Breakfast, Lunch, Dinner, Snack). The system aggregates cumulative daily calories, protein, carbohydrates, and fats in real-time, providing immediate visual feedback against the user's targeted nutritional goals.

5.4 Domain D: Thematic Recipe Collections & Bookmarking

Users can curate personal digital cookbooks by creating customized collections (e.g., 'Quick Weeknight Dinners', 'High-Protein Prep', 'Holiday Celebrations'). Collections feature customizable color themes, rich metadata, and one-click filtering.

6.0 MULTI-REGION NUTRITION INTELLIGENCE

Health Canada DRI Baseline & Global Target Matrix

Dietary advice online is often fraught with conflicting opinions and extreme macronutrient claims. Curation Kitchen anchors dietary health in peer-reviewed, public-health science by implementing a multi-region dietary reference engine.

THE HEALTH CANADA DRI DEFAULT MANDATE

Upon initial application launch, Curation Kitchen automatically defaults to Health Canada's Dietary Reference Intakes (DRIs). This establishes a balanced, evidence-based daily standard: 2,000 kcal, 56g Protein (0.8g/kg baseline), 225g Complex Carbohydrates (45-65% energy), 65g Healthy Fats (20-35% energy), and a strict 2,300mg Sodium moderation limit.

6.1 Multi-Region Target Matrix Specification

Users can freely toggle between international dietary frameworks or customize their daily targets based on specific athletic, metabolic, or therapeutic needs:

Region / Authority	Calories	Protein	Carbs	Fat	Sodium	Key Policy Basis
Health Canada (CA)	2,000 kcal	56 g	225 g	65 g	2,300 mg	Canada's Food Guide / DRI Baseline
USDA Guidelines (US)	2,000 kcal	50 g	275 g	78 g	2,300 mg	Dietary Guidelines for Americans
World Health Org (WHO)	2,000 kcal	50 g	250 g	66 g	2,000 mg	Global Chronic Disease Prevention
Custom Targets	User-Def.	User-Def.	User-Def.	User-Def.	User-Def.	Personalized Athletic / Medical Goals

6.2 Daily Nutrition Tracking & Progress Calculation

The Nutrition Tracker module features daily logging with dynamic date navigation. For any chosen day, the system computes total consumed nutrients from logged recipes and custom food entries, comparing them against the active target:

- **Progress Ratio Calculation:** Progress (%) = (Consumed Nutrient / Target Nutrient) × 100.
- **Safety Bounds & Visual Indicators:** Nutrient dials display progressive coloration: optimal green zones for balanced macronutrients, caution amber when approaching upper limits, and warning red when sodium exceeds maximum recommended thresholds.
- **Comprehensive Nutrient Logs:** Users can view granular meal logs, delete erroneous entries, or clear daily logs with full ACID persistence to the server backend.

7.0 ARTIFICIAL INTELLIGENCE PIPELINE

Google Gemini 3.6 Flash Integration & Reasoning Tasks

Artificial intelligence is integrated not as a superficial conversational novelty, but as a high-precision semantic parsing and calculation engine. Curation Kitchen employs the official Google Gen AI SDK (@google/genai) executing Gemini 3.6 Flash with 2.5 Flash fallbacks.

7.1 Intelligent AI Capabilities Breakdown

Endpoint / Route	Task Description	Output Schema
/api/recipes/import	URL web scraping & recipe parsing	Structured Recipe JSON with ingredients & steps
/api/gemini/parse-dictation	Spoken voice-to-recipe transcription	Parsed recipe title, quantities, steps
/api/gemini/estimate-nutrition	Predictive macro/micronutrient estimation	Calories, Protein, Carbs, Fat, Sodium
/api/gemini/modify-recipe	Dynamic substitutions & scaling	Transformed recipe with substituted ingredients
/api/gemini/lookup-upc	Barcode product categorization	Product title, brand, department category
/api/gemini/smart-shopping-list	Pantry-to-mealplan gap analysis	Consolidated missing items shopping list

7.2 Algorithmic Gap Analysis: The Smart Shopping List

One of the platform's most powerful AI features is the automated pantry gap analysis. When a user plans recipes for the week, the system queries both the planned meal ingredients and the active pantry ledger. The AI performs fuzzy semantic normalization, reconciles measurement units, and outputs a consolidated list containing only the items and quantities that must be purchased.

7.3 Safety, Temperature & Deterministic Outputs

All Gemini prompts enforce strict JSON schemas via system instructions and response formatting parameters. Temperature is configured to 0.2 to prioritize deterministic, fact-grounded outputs over creative hallucination. In the event of network timeouts or quota exhaustion, graceful fallbacks ensure users can continue manual recipe entry without interruption.

8.0 GROCERY E-COMMERCE & FULFILLMENT

The Missing Ingredient Pipeline & Affiliate Monetization

Curation Kitchen directly bridges the gap between digital planning and physical grocery delivery. Through its specialized Canadian e-commerce routing pipeline, missing ingredients can be ordered in a single tap across Canada's premier retail networks.

8.1 Fulfillment Partner Matrix

Retail Partner	Canadian Market Reach	Fulfillment Mode	Monetization Model
Instacart Canada	Loblaws, Superstore, Costco, Metro	1-2 Hour Express Delivery	Partner Affiliate Clicks/ Orders
Walmart Canada	Nationwide Urban & Suburban Stores	Same-Day Delivery & Pickup	Rakuten LinkShare Affiliate
Voilà by Sobeys	Ontario, Quebec, Atlantic Canada	Robotic Cold-Chain Direct	Impact Radius SubID Tracking

8.2 Deep Linking & Strict URI Encoding Engine

To ensure seamless cross-platform transitions without runtime crashes or URL truncation, all ingredient search queries pass through a centralized sanitization engine:

- **Whitespace Sanitization:** Trims leading, trailing, and duplicate spaces, stripping unprintable characters.
- **Percent-Encoding:** Strict RFC 3986 percent-encoding ensuring special characters, accents (e.g. 'crème fraîche'), and symbols are correctly transmitted as %20, %27, etc.
- **Affiliate Parameter Injection:** Dynamically appends tracking IDs (affiliate_id, utm_source, subid) to ensure conversions are accurately attributed.

9.0 SECURITY, OFFLINE RESILIENCE & PERFORMANCE

Hardening for Real-World Kitchen Use

9.1 API Key Isolation & Server-Side Security

In compliance with enterprise security guidelines, zero secret API tokens (Gemini, e-commerce credentials, payment secrets) are ever packaged into client-side web bundles or mobile APKs. All third-party interactions are securely mediated via Express backend proxy endpoints.

9.2 Offline Resilience & Local-First Operations

Cooking often occurs in environments with spotty Wi-Fi. Curation Kitchen incorporates robust offline defenses:

- **Client-Side IndexedDB & Local Storage:** Cached recipe data, active meal plans, and pantry states remain readable when disconnected.
- **Offline Banner & Visual Indicators:** The web and mobile clients monitor navigator.onLine and network state, displaying non-intrusive status indicators when offline.
- **Optimistic Mutation Queuing:** Pantry unit updates and checkbox toggles register instantaneously on the client, synchronizing with the server immediately upon reconnection.

9.3 Accessibility & Performance Benchmarks

- **Core Web Vitals:** Lighthouse performance scores consistently exceed 95, with First Contentful Paint (FCP) under 0.8s and zero Cumulative Layout Shift (CLS).
- **Contrast Standards:** All typography and interactive touch targets comply with WCAG 2.1 AA standards, with primary body text achieving 7.2:1 contrast against Buttercream backgrounds.
- **Touch Target Optimization:** Every interactive button and stepper on mobile exceeds the mandatory 44x44px minimum touch target size for comfortable one-handed use.

10.0 REST API SPECIFICATION & DATA SCHEMAS

Comprehensive Endpoint Reference

The Curation Kitchen backend exposes a clean, RESTful API surface. Below is a structured summary of primary system endpoints:

Method	Endpoint Route	Purpose & Payload	Response Format
GET	/api/recipes	Retrieve all saved recipes with filtering	Array<Recipe>
POST	/api/recipes	Create new recipe manually	Recipe object
PUT	/api/recipes/:id	Update recipe ingredients/instructions	Updated Recipe
DELETE	/api/recipes/:id	Remove recipe and dissociate collections	{ success: true }
POST	/api/recipes/import	AI scrape recipe from URL or text body	Parsed Recipe
GET	/api/pantry	Fetch complete pantry stock ledger	Array<PantryItem>
POST	/api/pantry	Add or increment ingredient in pantry	PantryItem object
PUT	/api/pantry/:id	Update unit quantity or department	Updated PantryItem
DELETE	/api/pantry/:id	Remove ingredient from pantry ledger	{ success: true }
GET	/api/mealplan	Fetch current 7-day meal plan schedule	MealPlan object
PATCH	/api/mealplan/slot	Assign or clear a specific meal slot	Updated MealPlan
GET	/api/nutrition/goals	Fetch current dietary target goals	NutritionGoals object
POST	/api/nutrition/goals	Update country baseline or custom goals	NutritionGoals object
GET	/api/nutrition/logs	Fetch daily nutrition log entries	Array<NutritionLog>
POST	/api/nutrition/logs	Log recipe or custom food item	NutritionLog object
POST	/api/gemini/lookup-upc	Query Open Food Facts & categorize	{ name, category }

11.0 ROADMAP & FUTURE RECOMMENDATIONS

Strategic Evolutions for Curation Kitchen

To further extend Curation Kitchen's market leadership as the premier digital culinary operating system, the following engineering initiatives are planned for subsequent development phases:

11.1 Computer Vision Fridge Inventory Scanning

Integrating Gemini Multimodal Vision to enable users to capture a photo of their open refrigerator or pantry shelf. The vision model detects vegetables, condiments, and dairy items in bulk, automatically updating pantry quantities with zero manual typing.

11.2 Multi-Household Collaborative Kitchen Sync

Enabling real-time collaborative syncing across family members using WebSockets or Firebase Firestore real-time listeners. When one household member purchases an item at the grocery store, the shared pantry updates instantaneously across all household devices.

11.3 IoT Smart Kitchen Appliance Integration

Establishing wireless Bluetooth and Matter protocol connectivity with smart kitchen scales, precision induction cooktops, and smart ovens to automate recipe timers and scale ingredient weights directly into the pantry ledger.

DOCUMENT CONCLUSION & SIGN-OFF

Curation Kitchen stands as a mature, multi-platform culinary operating system combining high-utility software engineering with evidence-based nutrition science. By anchoring dietary metrics to Health Canada DRIs, providing frictionless grocery delivery routing, and maintaining an elegant, tactile user experience, the platform delivers genuine culinary empowerment to cooks worldwide.

Approved by: Curation Kitchen System Architecture & Quality Assurance Board